

Bhagy Tholiya

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Education

IITRAM – Institute of Infrastructure Technology Research and Management Aug 2023 – Present
Bachelor of Technology in Computer Science Engineering | CGPA: 7.54 Ahmedabad, Gujarat

- Coursework: Machine Learning, Statistical Analysis, Database Management, Data Structures, Network Programming, Cryptography

Projects

Global Economic Intelligence Dashboard — Data Analytics & BI Reporting Tool [GitHub](#)

- Cleaned and analysed World Bank data across **195 countries** and 5 economic indicators to uncover key trends and patterns.
- Built an interactive Streamlit dashboard with Plotly maps and charts; used K-Means clustering to group countries into 3 economic profiles.
- Turned findings into clear, visual reports for non-technical stakeholders.

Real-Time Credit Card Fraud Detection — ML & Predictive Analytics Pipeline [GitHub](#)

- Cleaned and preprocessed **284,807 transactions**, fixed class imbalance with SMOTE, and engineered features to spot fraud patterns.
- Tuned the model threshold using business metrics, reaching a **ROC-AUC of 0.9807** with under 5ms inference via FastAPI.
- Built a reproducible 7-phase pipeline with drift monitoring and automated reporting.

Document Layout Analysis & Intelligent OCR — Deep Learning + NLP System [GitHub](#)

- Processed **85K+ PubLayNet** document images and analysed layout patterns across 5 structural classes to extract data from PDFs and scans.
- Added LLM-based summarisation using the Anthropic Claude API, cutting manual document review time by an estimated **40%**.

Chicken Disease Classification — Deep Learning for Predictive Diagnosis [GitHub](#)

- Labelled and preprocessed 390 fecal images, then trained a VGG16 model that reached **94.83% accuracy**.
- Deployed the model via a Flask REST API with a drag-and-drop interface, using DVC for a reproducible pipeline.

Virtual Internship Experience

Tata Group – Data Analytics Job Simulation (Forage) June 2026

- Explored customer data for Tata iQ’s Financial Services team using GenAI tools to spot risk patterns and build a simple model for predicting loan defaults.
- Designed an AI-assisted debt collection strategy that balanced automation with ethical and regulatory considerations.

Skills

Data Analytics	Data Cleaning & Preprocessing, EDA, Statistical Analysis, Hypothesis Testing, Feature Engineering, Business Intelligence, Reporting
Visualisation	Power BI, Plotly, Matplotlib, Seaborn, Streamlit, Google Sheets, Microsoft Excel
Python & Data	Python, Pandas, NumPy, Scikit-learn, SQL, MySQL, PostgreSQL, MongoDB, Jupyter Notebook
ML & AI	Machine Learning, Deep Learning, PyTorch, TensorFlow, XGBoost, LightGBM, LangChain, RAG, GenAI, Agentic AI
Tools	Git, GitHub, Docker, FastAPI, Flask, MLflow, DVC, Airflow, AWS Basics

Leadership & Activities

INFRA CYBERWISE-2026 — Infrastructure Cybersecurity Workshop Jan 2026
Event Organiser, Cyber Club IITRAM IITRAM, Ahmedabad

- Coordinated a two-day technical workshop for **100+ attendees**, presenting key findings and managing volunteers and logistics end-to-end.

CYBERWISE — Cybersecurity Workshop (Sponsored: CAWACH Kendra, Govt. of Gujarat) Nov 2025
Event Organiser, Cyber Club IITRAM IITRAM, Ahmedabad

- Co-organised a two-day workshop on cyber law, managing **150+ registrations** and earning an award for outstanding contribution.